

ให้

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}, a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, b = [3 \ 0 \ 8].$$

จงคำนวณผลคูณและผลบวกต่อไปนี้ หรือให้เหตุผลว่าทำไมถึงหาค่าไม่ได้ (Undefined)

1. $Ab^T + Bb^T, (A + B)b^T, bA, BA, B - B^T$
2. $A^T b, b^T B, (3A - 2B)^T a, a^T (3A - 2B)$
3. $ab - ba, -(4b)(7a), -28ba, 5abB$
4. $B^3, BC, (BC)^2, (BC)(BC)^T$
5. $IA, AI, A(I + A), A + IA$

คำตอบ

1. $\begin{bmatrix} \\ \\ \end{bmatrix}, \begin{bmatrix} \\ \\ \end{bmatrix}, \dots$

2.

3.

4.

Solutions and detailed explanations are provided on the subsequent pages.

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BY

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Problem 1

Given $A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$, $b = [3 \ 0 \ 8]$

Find 1) $Ab^T + Bb^T$

2) $(A + B)b^T$

3) bA

4) BA

5) $B - B^T$

Problem 1.1

$$Ab^T + Bb^T$$

First, find b^T

$$b = [3 \ 0 \ 8]$$

$$b^T = \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

Next, Compute $A \times b^T$

$$\begin{aligned} Ab^T &= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} (6 \times 3) + (-2 \times 0) + (2 \times 8) \\ (10 \times 3) + (-3 \times 0) + (1 \times 8) \\ (-10 \times 3) + (5 \times 0) + (1 \times 8) \end{bmatrix} = \begin{bmatrix} 34 \\ 38 \\ -22 \end{bmatrix} \\ &= \begin{bmatrix} 34 \\ 38 \\ -22 \end{bmatrix} \end{aligned}$$

Then, Compute $B \times b^T$

$$\begin{aligned} Bb^T &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} (9 \times 3) + (4 \times 0) + (-4 \times 8) \\ (4 \times 3) + (-7 \times 0) + (0 \times 8) \\ (-4 \times 3) + (0 \times 0) + (11 \times 8) \end{bmatrix} = \begin{bmatrix} -5 \\ 12 \\ 76 \end{bmatrix} \\ &= \begin{bmatrix} -5 \\ 12 \\ 76 \end{bmatrix} \end{aligned}$$

Thus,

$$\begin{aligned} Ab^T + Bb^T &= \begin{bmatrix} 34 \\ 38 \\ -22 \end{bmatrix} + \begin{bmatrix} -5 \\ 12 \\ 76 \end{bmatrix} \\ &= \begin{bmatrix} 34 + (-5) \\ 38 + 12 \\ -22 + 76 \end{bmatrix} = \begin{bmatrix} 29 \\ 50 \\ 54 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 29 \\ 50 \\ 54 \end{bmatrix}$$

Commented [PP1]: Answer 1.1

Problem 1.2

$$(A + B)b^T$$

First, find $(A + B)$

$$\begin{aligned} A + B &= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} + \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \\ &= \begin{bmatrix} 6+9 & (-2)+4 & 2+(-4) \\ 10+4 & (-3)+7 & 1+0 \\ -10+4 & 5+0 & 1+11 \end{bmatrix} = \begin{bmatrix} 15 & 2 & -2 \\ 14 & 4 & 1 \\ -6 & 5 & 12 \end{bmatrix} \\ &= \begin{bmatrix} 15 & 2 & -2 \\ 14 & 4 & 1 \\ -6 & 5 & 12 \end{bmatrix} \end{aligned}$$

Next, Evaluate $(A + B)b^T$

$$\begin{aligned} (A + B)b^T &= \begin{bmatrix} 15 & 2 & -2 \\ 14 & 4 & 1 \\ -6 & 5 & 12 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} (15 \times 3) + (2 \times 0) + (-2 \times 8) \\ (14 \times 3) + (4 \times 0) + (1 \times 8) \\ (-6 \times 3) + (5 \times 0) + (12 \times 8) \end{bmatrix} \\ &= \begin{bmatrix} 29 \\ 50 \\ 78 \end{bmatrix} \end{aligned}$$

Commented [PP2]: Answer 1.2

Problem 1.3

bA

Evaluate expression bA

$$\begin{aligned} bA &= \begin{bmatrix} 3 & 0 & 8 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (18 + 0 - 80) & (-6 + 0 + 40) & (6 + 0 + 8) \end{bmatrix} \\ &= \begin{bmatrix} -62 & 34 & 14 \end{bmatrix} \end{aligned}$$

Commented [PP3]: Answer 1.3

Problem 1.4

BA

Evaluate expression BA

$$\begin{aligned} BA &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 54 + 40 + 40 & -18 - 12 - 20 & 18 + 4 - 4 \\ 24 + 70 + 0 & -8 - 21 + 0 & 8 + 7 + 0 \\ -24 + 0 - 110 & 8 + 0 + 55 & -8 + 0 + 11 \end{bmatrix} \\ &= \begin{bmatrix} 134 & -50 & 18 \\ 94 & -29 & 15 \\ -134 & 63 & 3 \end{bmatrix} \end{aligned}$$

Commented [PP4]: Answer 1.4

Problem 1.5

$$B - B^T$$

Calculate $B - B^T$

$$\begin{aligned} B - B^T &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} - \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \\ &= \begin{bmatrix} 9-9 & 4-4 & -4-(-4) \\ 4-4 & 7-7 & 0-0 \\ -4-(-4) & 0-0 & 11-11 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Commented [PP5]: Answer 1.5

Problem 2

Given

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$
$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, \quad b = [3 \quad 0 \quad 8]$$

Find

- 1) $A^T b$
- 2) $b^T B$
- 3) $(3A - 2B)^T a$
- 4) $a^T (3A - 2B)$

Problem 2.1

$$A^T b$$

First, find A^T

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$
$$A^T = \begin{bmatrix} 6 & 10 & -10 \\ -2 & -3 & 5 \\ 2 & 1 & 1 \end{bmatrix}$$

Next, Compute $A^T b$

A^T is a 3×3 matrix, and b is a 1×3 matrix.

Since the inner dimensions (3 and 1) do not match, the multiplication is *Undefined*.

Commented [PP6]: Answer 2.1

Problem 2.2

$$b^T B$$

First, find b^T

$$b = [3 \quad 0 \quad 8]$$
$$b^T = \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

Next, Compute $b^T B$

b^T is a 3×1 matrix, and B is a 3×3 matrix.

Since the inner dimensions (1 and 3) do not match, the multiplication is *Undefined*.

Commented [PP7]: 2.2

Problem 2.3

$$(3A - 2B)^T a$$

First, find $3A - 2B$

$$\begin{aligned} 3A &= 3 \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 18 & -6 & 6 \\ 30 & -9 & 3 \\ -30 & 15 & 3 \end{bmatrix} \\ 2B &= 2 \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 18 & 8 & -8 \\ 8 & 14 & 0 \\ -8 & 0 & 22 \end{bmatrix} \\ 3A - 2B &= \begin{bmatrix} 18 - 18 & -6 - 8 & 6 - (-8) \\ 30 - 8 & -9 - 14 & 3 - 0 \\ -30 - (-8) & 15 - 0 & 3 - 22 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix} \end{aligned}$$

Next, find $(3A - 2B)^T$

$$(3A - 2B)^T = \begin{bmatrix} 0 & 22 & -22 \\ -14 & -23 & 15 \\ 14 & 3 & -19 \end{bmatrix}$$

Then, Compute $(3A - 2B)^T a$

$$(3A - 2B)^T a = \begin{bmatrix} 0 & 22 & -22 \\ -14 & -23 & 15 \\ 14 & 3 & -19 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{aligned}
 &= \begin{bmatrix} (0 \times 5) + (22 \times 1) + (-22 \times 2) \\ (-14 \times 5) + (-23 \times 1) + (15 \times 2) \\ (14 \times 5) + (3 \times 1) + (-19 \times 2) \end{bmatrix} \\
 &= \begin{bmatrix} 0 + 22 - 44 \\ -70 - 23 + 30 \\ 70 + 3 - 38 \end{bmatrix}
 \end{aligned}$$

$$= \begin{bmatrix} -22 \\ -63 \\ 35 \end{bmatrix}$$

Problem 2.4

$$a^T(3A - 2B)$$

First, find a^T

$$\begin{aligned}
 a &= \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\
 a^T &= [5 \quad 1 \quad 2]
 \end{aligned}$$

We already have $3A - 2B = \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix}$

Next, Compute $a^T(3A - 2B)$

$$\begin{aligned}
 a^T(3A - 2B) &= [5 \quad 1 \quad 2] \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix} \\
 &= [(0 + 22 - 44) \quad (-70 - 23 + 30) \quad (70 + 3 - 38)] \\
 &= \begin{bmatrix} -22 & -63 & 35 \end{bmatrix}
 \end{aligned}$$

Commented [PP8]: Answer 2.3

Commented [PP9]: Answer 2.4

Problem 3

Given

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$
$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, \quad b = [3 \quad 0 \quad 8]$$

Find

- 1) $ab - ba$
- 2) $-(4b)(7a)$
- 3) $-28ba$
- 4) $5abB$

Problem 3.1

$$ab - ba$$

First, compute ab

$$\begin{aligned} ab &= \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} [3 \quad 0 \quad 8] \\ &= \begin{bmatrix} (5 \times 3) & (5 \times 0) & (5 \times 8) \\ (1 \times 3) & (1 \times 0) & (1 \times 8) \\ (2 \times 3) & (2 \times 0) & (2 \times 8) \end{bmatrix} \\ &= \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix} \end{aligned}$$

Next, compute ba

$$\begin{aligned}
 ba &= [3 \quad 0 \quad 8] \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\
 &= [(3 \times 5) + (0 \times 1) + (8 \times 2)] \\
 &= [15 + 0 + 16] \\
 &= [31]
 \end{aligned}$$

Then, calculate $ab - ba$

Since ab is a 3×3 matrix and ba is a 1×1 matrix, they have different dimensions.

Therefore, subtraction is **Undefined**.

Commented [PP10]: Answer 3.1

Problem 3.2

$$-(4b)(7a)$$

First, find $4b$ and $7a$

$$\begin{aligned}
 4b &= 4[3 \quad 0 \quad 8] \\
 &= [12 \quad 0 \quad 32]
 \end{aligned}$$

$$\begin{aligned}
 7a &= 7 \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\
 &= \begin{bmatrix} 35 \\ 7 \\ 14 \end{bmatrix}
 \end{aligned}$$

Then, compute $-(4b)(7a)$

$$\begin{aligned}
 -(4b)(7a) &= -[12 \quad 0 \quad 32] \begin{bmatrix} 35 \\ 7 \\ 14 \end{bmatrix} \\
 &= -[(12 \times 35) + (0 \times 7) + (32 \times 14)] \\
 &= -[420 + 0 + 448] \\
 &= [-868]
 \end{aligned}$$

Therefore,

$$-(4b)(7a) = [-868]$$

Commented [PP11]: Answer 3.2

Problem 3.3

$$-28ba$$

We already have $ba = [31]$

Compute $-28ba$

$$-28ba = -28[31]$$

$$= [-868]$$

Commented [PP12]: Answer 3.3

Problem 3.4

$$5abB$$

$$\text{We already have } ab = \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix}$$

First, find $5ab$

$$5ab = 5 \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix}$$

$$a = \begin{bmatrix} 75 & 0 & 200 \\ 15 & 0 & 40 \\ 30 & 0 & 80 \end{bmatrix}$$

Next, compute $(5ab)B$

$$5abB = \begin{bmatrix} 75 & 0 & 200 \\ 15 & 0 & 40 \\ 30 & 0 & 80 \end{bmatrix} \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} (75 \times 9) + (0 \times 4) + (200 \times -4) & (75 \times 4) + (0 \times 7) + (200 \times 0) & (75 \times -4) + (0 \times 0) + (200 \times 11) \\ (15 \times 9) + (0 \times 4) + (40 \times -4) & (15 \times 4) + (0 \times 7) + (40 \times 0) & (15 \times -4) + (0 \times 0) + (40 \times 11) \\ (30 \times 9) + (0 \times 4) + (80 \times -4) & (30 \times 4) + (0 \times 7) + (80 \times 0) & (30 \times -4) + (0 \times 0) + (80 \times 11) \end{bmatrix}$$

$$= \begin{bmatrix} 675 + 0 - 800 & 300 + 0 + 0 & -300 + 0 + 2200 \\ 135 + 0 - 160 & 60 + 0 + 0 & -60 + 0 + 440 \\ 270 + 0 - 320 & 120 + 0 + 0 & -120 + 0 + 880 \end{bmatrix}$$

$$= \begin{bmatrix} -125 & 300 & 1900 \\ -25 & 60 & 380 \\ -50 & 120 & 760 \end{bmatrix}$$

Commented [PP13]: Answer 3.4

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Problem 4

Given

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}, C = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

Find

- 1) B^3
- 2) BC
- 3) $(BC)^2$
- 4) $(BC)(BC)^T$

Problem 4.1

$$B^3$$

First, compute B^2

$$\begin{aligned} B^2 &= B \times B \\ &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \\ &= \begin{bmatrix} (9 \times 9) + (4 \times 4) + (-4 \times -4) & (9 \times 4) + (4 \times 7) + (-4 \times 0) & (9 \times -4) + (4 \times 0) + (-4 \times 11) \\ (4 \times 9) + (7 \times 4) + (0 \times -4) & (4 \times 4) + (7 \times 7) + (0 \times 0) & (4 \times -4) + (7 \times 0) + (0 \times 11) \\ (-4 \times 9) + (0 \times 4) + (11 \times -4) & (-4 \times 4) + (0 \times 7) + (11 \times 0) & (-4 \times -4) + (0 \times 0) + (11 \times 11) \end{bmatrix} \\ &= \begin{bmatrix} 81 + 16 + 16 & 36 + 28 + 0 & -36 + 0 - 44 \\ 36 + 28 + 0 & 16 + 49 + 0 & -16 + 0 + 0 \\ -36 + 0 - 44 & -16 + 0 + 0 & 16 + 0 + 121 \end{bmatrix} \\ &= \begin{bmatrix} 113 & 64 & -80 \\ 64 & 65 & -16 \\ -80 & -16 & 137 \end{bmatrix} \end{aligned}$$

Next, compute $B^3 = B^2 B$

$$\begin{aligned}
 B^3 &= \begin{bmatrix} 113 & 64 & -80 \\ 64 & 65 & -16 \\ -80 & -16 & 137 \end{bmatrix} \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \\
 &= \begin{bmatrix} (1017 + 256 + 320) & (452 + 448 + 0) & (-452 + 0 - 880) \\ (576 + 260 + 64) & (256 + 455 + 0) & (-256 + 0 - 176) \\ (-720 - 64 - 548) & (-320 - 112 + 0) & (320 + 0 + 1507) \end{bmatrix}
 \end{aligned}$$

$$= \begin{bmatrix} 1593 & 900 & -1332 \\ 900 & 711 & -432 \\ -1332 & -432 & 1827 \end{bmatrix}$$

Commented [PP14]: Answer 4.1

Problem 4.2

BC

Compute $B \times C$

$$\begin{aligned}
 BC &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} (9 \times 3) + (4 \times 0) + (-4 \times 4) & (9 \times 1) + (4 \times -2) + (-4 \times 0) \\ (4 \times 3) + (7 \times 0) + (0 \times 4) & (4 \times 1) + (7 \times -2) + (0 \times 0) \\ (-4 \times 3) + (0 \times 0) + (11 \times 4) & (-4 \times 1) + (0 \times -2) + (11 \times 0) \end{bmatrix} \\
 &= \begin{bmatrix} 27 + 0 - 16 & 9 - 8 + 0 \\ 12 + 0 + 0 & 4 - 14 + 0 \\ -12 + 0 + 44 & -4 + 0 + 0 \end{bmatrix}
 \end{aligned}$$

$$= \begin{bmatrix} 11 & 1 \\ 12 & -10 \\ 32 & -4 \end{bmatrix}$$

Commented [PP15]: Answer 4.2

Problem 4.3

$$(BC)^2$$

$(BC)^2 = (BC)(BC)$ is a 3×2 matrix. Since we are multiplying a 3×2 matrix by another 3×2 matrix, the inner dimensions (2 and 3) do not match. Therefore, the multiplication is *Undefined*.

Commented [PP16]: Answer 4.3

Problem 4.4

$$(BC)(BC)^T$$

We already have $BC = \begin{bmatrix} 11 & 1 \\ 12 & -10 \\ 32 & -4 \end{bmatrix}$

First, find $(BC)^T$

$$(BC)^T = \begin{bmatrix} 11 & 12 & 32 \\ 1 & -10 & -4 \end{bmatrix}$$

Next, Compute $(BC)(BC)^T$

$$\begin{aligned} (BC)(BC)^T &= \begin{bmatrix} 11 & 1 \\ 12 & -10 \\ 32 & -4 \end{bmatrix} \begin{bmatrix} 11 & 12 & 32 \\ 1 & -10 & -4 \end{bmatrix} \\ &= \begin{bmatrix} (11 \times 11) + (1 \times 1) & (11 \times 12) + (1 \times -10) & (11 \times 32) + (1 \times -4) \\ (12 \times 11) + (-10 \times 1) & (12 \times 12) + (-10 \times -10) & (12 \times 32) + (-10 \times -4) \\ (32 \times 11) + (-4 \times 1) & (32 \times 12) + (-4 \times -10) & (32 \times 32) + (-4 \times -4) \end{bmatrix} \\ &= \begin{bmatrix} 121 + 1 & 132 - 10 & 352 - 4 \\ 132 - 10 & 144 + 100 & 384 + 40 \\ 352 - 4 & 384 + 40 & 1024 + 16 \end{bmatrix} \\ &= \begin{bmatrix} 122 & 122 & 348 \\ 122 & 244 & 424 \\ 348 & 424 & 1040 \end{bmatrix} \end{aligned}$$

Commented [PP17]: Answer 4.4

Problem 5

Given

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Assuming I is the identity matrix.

Find

- 1) IA
- 2) AI
- 3) $A(I + A)$
- 4) $A + IA$

Problem 5.1

IA

Since I is the identity matrix, multiplying any matrix by I yields the matrix itself.

$$\begin{aligned} IA &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \end{aligned}$$

Commented [PP18]: Answer 5.1

Problem 5.2

$$AI$$

Similarly, multiplying A by I yields the matrix A itself.

$$AI = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

Commented [PP19]: Answer 5.2

Problem 5.3

$$A(I + A)$$

First, find $I + A$

$$\begin{aligned} I + A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+6 & 0+(-2) & 0+2 \\ 0+10 & 1+(-3) & 0+1 \\ 0+(-10) & 0+5 & 1+1 \end{bmatrix} \\ &= \begin{bmatrix} 7 & -2 & 2 \\ 10 & -2 & 1 \\ -10 & 5 & 2 \end{bmatrix} \end{aligned}$$

Next, Compute $A(I + A)$

$$\begin{aligned} A(I + A) &= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \begin{bmatrix} 7 & -2 & 2 \\ 10 & -2 & 1 \\ -10 & 5 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (42 - 20 - 20) & (-12 + 4 + 10) & (12 - 2 + 4) \\ (70 - 30 - 10) & (-20 + 6 + 5) & (20 - 3 + 2) \\ (-70 + 50 - 10) & (20 - 10 + 5) & (-20 + 5 + 2) \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2 & 14 \\ 30 & -9 & 19 \\ -30 & 15 & -13 \end{bmatrix} \end{aligned}$$

Commented [PP20]: Answer 5.3

Problem 5.4

$$A + IA$$

Since $IA = A$, we have $A + IA = A + A$

First, we already have $IA = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$

Then, compute $A + IA$

$$\begin{aligned} A + IA &= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} + \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 6+6 & -2-2 & 2+2 \\ 10+10 & -3-3 & 1+1 \\ -10-10 & 5+5 & 1+1 \end{bmatrix} \\ &= \begin{bmatrix} 12 & -4 & 4 \\ 20 & -6 & 2 \\ -20 & 10 & 2 \end{bmatrix} \end{aligned}$$

Commented [PP21]: Answer 5.4